

Student Mental Health Monitoring System

SE LAB Experiment-04 BATCH-D

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1 SYSTEM MODELS

1.1 Structured Use Case Model

ID	Primary Actor	Relation Type	Description
UC-01	Student	Core	Register / Login into the system.
UC-02	Student	Core	Fill standardized mental health screening forms.
UC-04	Student	Core	View mental health status and risk level.
UC-11	System	Core	Analyze academic and behavioral patterns.
UC-12	System	«include» UC-11	Detect mental health risk levels.
UC-15	Counselor	Core	View at-risk students.
UC-18	Counselor	Core	Schedule counseling sessions.
UC-21	Admin	Core	Manage users (students and counselors).
UC-22	Admin	Core	Monitor overall institutional mental health trends.
UC-29	IT Security Officer	Core	Manage data access and audit logs.

2 ANALYSIS MODEL (ENTITY-BOUNDARY-CONTROL)

2.1 UC-01: Register / Login

Entity Objects:

- **UserCredentials** — Stores the student's login credentials used for authentication.

Boundary Objects:

- **LoginInterface** — UI screen that captures and submits user login details.

Control Objects:

- **AuthenticationController** — Validates credentials and manages the login/logout flow.
- **SessionManagementController** — Creates and maintains the active user session after login.

2.2 UC-02: Fill Screening Form

Entity Objects:

- **ScreeningResponse** — Captures and persists the student's answers from the screening form.

Boundary Objects:

- **ScreeningFormInterface** — UI form through which the student enters mental health responses.

Control Objects:

- **ScreeningController** — Orchestrates form validation and submission of screening data.
- **DataCollectionController** — Collects and routes submitted responses for storage.

2.3 UC-04: View Mental Health Status

Entity Objects:

- **MentalHealthRecord** — Holds the student's scored screening history and current risk status.

Boundary Objects:

- **StudentDashboardUI** — Displays the student's current mental health status and risk level.

Control Objects:

- **RiskAnalysisController** — Retrieves and processes risk data for display on the dashboard.
- **DashboardController** — Coordinates data loading and rendering for the student view.

2.4 UC-11: Analyze Patterns

Entity Objects:

- **BehavioralData** — Aggregates academic performance, attendance, and mood logs for analysis.

Boundary Objects:

- **SystemAnalyticsInterface** — Internal interface that triggers and displays pattern analysis results.

Control Objects:

- **PatternAnalysisController** — Runs algorithms to identify trends across academic and behavioral data.
- **DataProcessingController** — Preprocesses and normalizes raw data prior to analysis.

2.5 UC-12: Detect Risk Levels

Entity Objects:

- **RiskAssessment** — Stores the classified risk level (low/medium/high) assigned to each student.

Boundary Objects:

- **SystemAnalyticsInterface** — Internal interface that receives pattern results and outputs risk levels.

Control Objects:

- **RiskAnalysisController** — Applies risk thresholds to scored data and updates risk status.
- **ClassificationController** — Categorizes students into risk tiers based on computed scores.

2.6 UC-15: View At-Risk Students

Entity Objects:

- **RiskAssessment** — Provides the current risk classification of each student for counselor review.

Boundary Objects:

- **CounselorDashboardUI** — Displays a filtered list of students currently flagged as at-risk.

Control Objects:

- **RiskMonitoringController** — Queries and filters students based on their risk assessment status.
- **ReportController** — Formats and delivers risk summary reports for counselor review.

2.7 UC-18: Schedule Counseling

Entity Objects:

- **CounselingSession** — Represents a booked appointment with date, time, and participant details.

Boundary Objects:

- **CounselingScheduleInterface** — UI for selecting time slots and confirming session bookings.

Control Objects:

- **CounselingSessionController** — Manages creation, modification, and cancellation of sessions.
- **NotificationController** — Sends confirmation and reminder alerts to students and counselors.

2.8 UC-21: Manage Users

Entity Objects:

- **UserProfile** — Represents any system user (student or counselor) managed by the admin.

Boundary Objects:

- **AdminPanelUI** — Interface through which the admin manages all user accounts and roles.

Control Objects:

- **UserManagementController** — Handles CRUD operations for all system user accounts.
- **RoleAssignmentController** — Assigns and updates access roles for students and counselors.

2.9 UC-22: Monitor Overall Trends

Entity Objects:

- **DashboardReport** — Aggregated summary of institution-wide mental health trends for admin review.

Boundary Objects:

- **AdminDashboardUI** — Displays graphs and statistics representing overall mental health trends.

Control Objects:

- **TrendAnalysisController** — Aggregates and analyzes records to produce institution-wide summaries.
- **DashboardController** — Coordinates data retrieval and rendering for the admin dashboard view.

2.10 UC-29: Manage Data Access

Entity Objects:

- **AuditLog** — Records all data access and modification events for compliance and review.

Boundary Objects:

- **SecurityConfigurationInterface** — UI for the IT officer to configure policies and review access logs.

Control Objects:

- **PrivacyComplianceController** — Ensures data handling adheres to privacy regulations and policies.
- **AccessControlController** — Enforces access permissions and logs any unauthorized attempts.